

Coastal Dunes: Succession and Nitrogen Availability on Functional Plant Groups

Abstract

Ecological succession occurs from changes in soil composition and microclimate, leading to changes in plant vegetation. This study examines how variations in plant diversity, abundance, and species richness over successional time can be explained by environmental factors along ocean dunes in Douglas County, Oregon. Data were collected along a transect from the beach to the forest, where plant species were identified, categorized into functional groups, and percent cover was calculated. Nitrogen content was measured using soil samples. Results showed significant differences in species richness across successional stages, with higher richness in later stages. Woody plants thrived in later stages of succession as herbaceous plants thrived the most in the foredune. Beachgrass displayed negative correlation with nitrogen content due to them being nitrogen fixers. The woody functional group displayed a positive correlation with nitrogen content as they rely on nitrogen for growth.

Introduction

Ecological succession is defined as the process by which the community replaces or succeeds one another from modification of the physical environment over time (Odum 1969). Coastal dunes are particularly useful for studying ecological succession as they are characterized by their highly dynamic topography, and undergoing constant modifications of soil content and microclimate through space and time, resulting in changes in vegetation composition. As dunes develop, they progress through distinct successional stages, beginning with pioneer species stabilizing shifting sands, followed by the establishment of grasses and shrubs, and leading to mature forests farther inland. Sand dunes require an abundant supply of sand, strong winds, and obstacles to trap sand (through plants). Beaches are shaped by the tides and ocean currents as they transport sand inland through high winds. Oregon sand dunes contain high volumes of *Ammophila arenaria* (European beachgrass), an introduced species to the area, which creates high dunes through it capturing sand particles in the foredune. Just beyond the foredune lies the deflation plain, which emerges when winds strip sand from the east, allowing for vast plant vegetation growth from exposure of the water table (Schultz 1990). *Ammophila arenaria* has decreased the space of the open dunes, resulting in rapid succession of shrubs and forests (Wiedemann and Pickart 1996). Soil nutrients such as nitrogen also change through space, and influence vegetation composition and subsequent successions. Sand easily loses

nutrients due to rain, wind, and the ocean, as it has limited water-holding capacity, which restricts mineralization and nitrification, resulting in decreased inorganic nitrogen content, which is essential for plant growth and function (Kachi, 1983). With the presence of beachgrass, some stabilization occurs, in regards to trapping sand as well as fixing nitrogen over time to increase soil nutrient content. Exploring plant distribution and nutrient availability between these ecosystems will be crucial for understanding ecological succession. This study aimed to assess plant functional group distributions, plants that have similar characteristics or fill a particular niche within an ecosystem, along a successional gradient from beach to forest. Functional groups we will be looking at include: trees which are woody plants with tall trunks (red alder, sitka spruce, spore pine, doug fir), shrubs which are herbaceous plants (bearberry, scotch broom, garrya, salal, wax myrtle, rhododendron, willow, evergreen), forbes, graminoids, and ferns. We predicted that there will be higher species richness over successional time because the accumulation of soil nutrients and stabilization of environmental conditions through increased vegetation will support greater diversity of plant functional groups, leading to increased species richness. We predict that woody organisms will become more dominant while herbaceous organisms will decrease with successional time because of differences in their tolerance to salinity and shade, with woody species better adapted to increased stable and shaded environments (Olf, 1993). We predict that species richness will increase with successional time, because nitrogen content increases, which allows a wider variety of plants to thrive and grow more efficiently as it has a necessary nutrient for growth (Olf, 1993).

Methods

We collected data on plant species from the dunes at the Oregon Dunes Day Use Area in Douglas County, Oregon (43°43'27" North, 124°10'39"West) located in the Siuslaw National Forest. This national forest spans over 630,000 along the Oregon Coast Range from Tillamook to Coos Bay. This diverse landscape transitions from open sand dunes to dense forests, supporting a rich variety of organisms. This national park is described as Pacific maritime as it has mild winters (with average temperatures between -1 °C and 5 °C), cool/dry summers (with average temperatures in between 15 °C and 10 °C), and high annual precipitation, moderated by the influence of the Pacific Ocean (USDA, 2025). The area has an average precipitation of 47.45 inches of precipitation annually, though it can reach 100 inches (USDA, 2025). We created a primary transect along a successional gradient from the

forest to the beach. The transect consisted of 30 points at every 50 m from the beginning of the trail to the beach. At each point, we collected 5 meters of data perpendicular to the left and right side of the trail. We assigned 20 points at 0.25 m intervals and recorded all plant species intercepted within 1 cm of the point. We grouped species by functional groups and identified coverage of each point on the transect. To calculate the percent cover, the total number of intercepts for each species or substrate was divided by the total number of points sampled and multiplied by 100. The percent cover was then averaged across all transects within a vegetation zone to provide an overall estimate for each species or substrate. Cover by functional group was determined by summing the cover for relevant species. We collected the elevation and nitrogen levels at each central point along the transect. To collect the nitrogen content, soil samples were collected and dried at 60°C for 48 hours and analyzed using a Costech Analytical Technologies 4010 elemental combustion analyzer (Valencia, CA, USA). Samples were collected at each of the flags along the central transect. Some confounding variables include disturbances and human activity. We compared and assessed for statistical significance of functional groups through time using ANOVA, Tukey's Post Hoc Test, and linear regression using Excel (16.95) and Python (3.11.5).

Results

There were significant differences among species richness over successional groups ($df = 4$, $F_s = 52.8$, $p < 0.001$) (Figure 1A). There were no significant differences between beach (0.000 ± 0.000) and open dunes (0.58 ± 0.42). There also was no significant differences between the foredune (3.63 ± 0.89), deflation plane (4.33 ± 0.61), and forest (4.06 ± 0.46) (Figure 1A). There were significant differences between these two groups of organisms ($p < 0.001$) (Figure 1A). There were significant differences among successional age for woody and herbaceous organisms ($df = 9$, $F_s = 70.1$, $p < 0.05$) (Figure 1B). Between groups, there were significant differences between woody organisms in the deflation plane (21.92 ± 3.24) and forest (25.56 ± 2.93) compared to all other groups ($p < 0.05$). There was no significant difference between both groups at the beach and open dunes. Between successional herbaceous groups, there was only significant difference at the foredunes (8.75 ± 2.85 , $p < 0.001$) (Figure 1B).

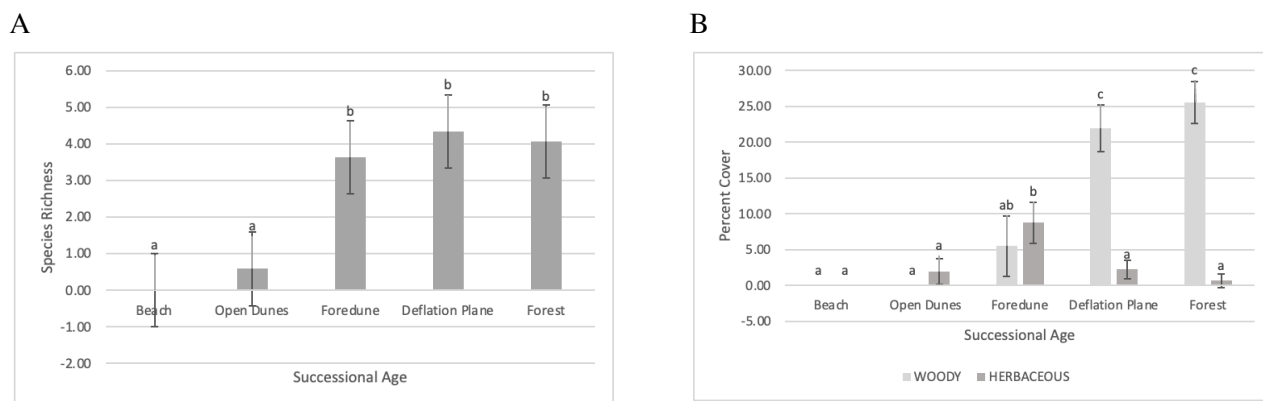


Figure 1. Summary of percent cover of Species Richness (A) and Percent cover of woody and herbaceous organisms (B) across successional age (df = 4, n = 68). Error bars represent 95% confidence intervals. A: Species richness had significant differences between beach and open dunes when compared to other succession groups ($p < 0.05$). B: There was a high significant difference between woody and herbaceous organisms at the deflation plane and forest ($p < 0.001$).

A linear regression model was fitted to examine the relationship between nitrogen content and beachgrass and woody percent cover. The linear regression model for beachgrass was statistically significant and resulted in a slope of -2.094 with an intercept of 2.378 ($R^2 = 0.088$, $p < 0.05$) (Figure 2C). The linear regression model for woody percent cover was statistically significant and resulted in a slope of 0.02 with an intercept of 0.03 ($R^2 = 0.408$, $p < 0.001$) (Figure 2D).

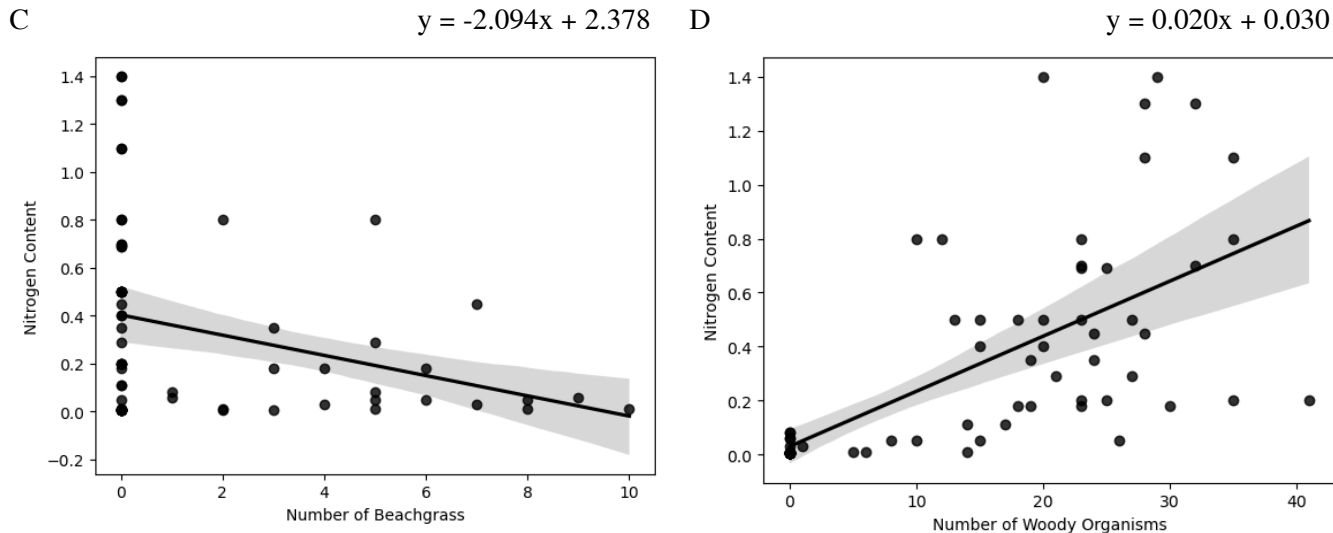


Figure 2. Summary of the effect of beachgrass (C) and woody organisms (D) on nitrogen content (n = 76). C: Regression for the number of beachgrass shows a significant negative correlation ($R^2 = 0.088$, $p < 0.05$). D: Regression for the number of woody organisms shows a significant positive correlation ($R^2 = 0.408$, $p < 0.001$). Shaded regions represent 95% confidence intervals and solid lines represent the fitted linear model.

Discussion

Our findings support that variations in plant diversity, abundance, and species richness over successional time can be explained by environmental factors influencing their distribution.

A long-term study conducted on the Wadden Sea island of Schiermonnikoog (Netherlands) examined the vegetation structure and soil development of dunes over 18 years. Researchers found that nitrogen limited above-ground biomass production by drying clippings of organisms found in different areas throughout the dunes. They found that nitrogen limited above-ground biomass production and that as succession progresses, nitrogen accumulation increases, which is accompanied by increased biomass, decreased light penetration to the soil surface, decreased root/shoot ratio, increasing dominance of tall species, and decreasing abundance of small, short lived species (Olf, 1993).

Their results showed that organisms in poor nitrogen conditions focused on root growth, while organisms in rich nitrogen areas focused on shoot growth. In areas with limited nitrogen, *Elymus farctus* and *Ammophila arenaria* were mostly present, as both species can fix their own nitrogen and thrive in spaces without nitrogen (Pavlik, 1983). These particular organisms also invest in longer root development, explaining why they have less above-ground biomass. These findings aligned with the data collected in this study, which showed negative correlation between introduced and native species of beachgrass when compared to nitrogen content (Figure 2C).

Other organisms depend on inorganic nitrogen to produce chlorophyll and proteins necessary for photosynthesis and growth, preventing them from thriving in areas without it (Kachi, 1983). This aligns with the results of a positive correlation between woody content and nitrogen content as these organisms rely on nitrogen content in soil for growth and invest more of their energy into their shoots, creating a higher above-ground biomass. In addition, because woody organisms focus on shoot growth, there is increased light interception resulting in decreased light to soil surfaces, which favors shade tolerant organisms; this identifies why woody organisms increase with successional age (Figure 1B). As shade-tolerant organisms thrive under these conditions, the abundance of non-shade-tolerant species declines, which helps explain the dynamic observed of a decrease in herbaceous plant populations as succession progresses (Figure 1B).

Species richness increases with successional age, exhibiting a significant increase at the foredunes, after which it stabilizes at a consistent level (Figure 1A). A study conducted in St George Island, Florida, examined how changes in elevation levels of different habitats (i.e. foredune, interdune, backdune) impacted species richness and plant composition. In the foredune, species richness was initially low, but rapidly increased. This environment

typically experiences the most changes in elevation, but due to the presence of *Ammophila arenaria*, the dunes have stabilized, resulting in creation of microhabitats that can support a greater diversity of organisms than it could have previously (Miller, 2015; Stalter, 2024). In addition, this may result from decreased salinity as this can affect the abundance of organisms as many cannot tolerate these conditions (Olff, 1993). The interdune and backdune experience less disturbance than successional stages before it, which results in higher species richness. It is important to note that the diversity of these organisms differs in these successional stages as the microhabitats fill specific ecological niches for organisms. The foredune still experiences strong winds, which results in herbaceous organisms and shorter woody organisms thriving (Figure 1B). At the deflation plain, the water table is exposed, so organisms that require more constant water thrive (woody organisms) (Figure 1B). In the forest, there is less disturbance and more nutrients, so a wider variety of plants can thrive resulting in increased species richness.

One limitation of this study could be that measurements were taken over one day during the winter season, when wind patterns differ from summer. A different study conducted on the Oregon dunes found that composition of vegetation on the dunes changed rapidly with shifting of sand from wind pattern changes from season to season (Kumler 1963). This would result in potentially shifting boundaries for successional stages, as dunes stabilize in different areas and plant vegetation changes over time, which could not accurately reflect the long-term patterns of dune development and vegetation dynamics with nitrogen availability. In addition, because the transect was conducted along a human built trail, there could be changes in plant vegetation from human interaction. Future research should focus on how soil characteristics beyond nitrogen content impact species richness and plant composition in the dunes. Understanding how soil factors impact plant growth and succession is important for conservation of the dunes as *Ammophila arenaria* has changed the structure of the dunes.

In conclusion, this study explores how nitrogen availability and environmental factors impact plant diversity, abundance, and species richness in dune ecosystems. Woody organisms rely on nitrogen rich soil for growth as beachgrass and other herbaceous organisms can thrive in nitrogen-poor areas as they are nitrogen fixers. Herbaceous organisms decrease over successional time from competition. In addition, species richness increases with successional age, as it provides more stable environments through increased nutrient supply and less disturbance.

Bibliography

- Wiedemann, A. M., L. R. J. Dennis, and F. H. Smith. 1974. Plants of the Oregon coastal dunes. Oregon State University Bookstore, Corvallis, Oregon.
- Wiedemann, A. M. and A. Pickart. 1996. The *Ammophila* problem on the Northwest Coast of North America. *Landscape and Urban Planning* 34:287-299.
- Kumler, M. L. 1968. Succession of plants native to sand dunes of Oregon coast. *American Journal of Botany* 55:717.
- Odum, Eugene P. "The strategy of Ecosystem Development." *Science*, vol. 164, no. 3877, 18 Apr. 1969, 262-270.
- Schultz, S. T. 1990. The Northwest coast: A natural history. Timber Press, Portland, Oregon.
- Walker, L. R., & del Moral, R. (2003). *Primary succession and ecosystem rehabilitation*. Cambridge University Press.
- Olf, H., et al. 1993. Species Dynamics and Nutrient Accumulation During Early Primary Succession in Coastal Sand Dunes. *Journal of Ecology* 81:693–706
- "Siuslaw National Forest - about the Area." *Forest Service National Website*,
www.fs.usda.gov/detail/siuslaw/about-forest/about-area/?cid=fsbdev7_007333.
- Pavlik BM. 1983. Nutrient and productivity relations of the dune grasses *Ammophila arenaria* and *Elymus mollis*. Blade photosynthesis and nitrogen use efficiency in the laboratory and field. *Oecologia* 57(1-2):227-232.
- Kachi, N., and T. Hirose. 1983. "Limiting Nutrients for Plant Growth in Coastal Sand Dune Soils." *Journal of Ecology* 71:937–44.
- Miller TE. 2015. Effects of disturbance on vegetation by sand accretion and erosion across coastal dune habitats on a barrier island. *AoB Plants* 12-7.